

|           |                                                                         |           |
|-----------|-------------------------------------------------------------------------|-----------|
|           |                                                                         |           |
| 10(a)     | the steady current<br><b>or</b><br>the direct current                   | <b>M1</b> |
|           | that produces the same heating effect (as the alternating current)      | <b>A1</b> |
| 10(b)(i)  | peak current = 2.6 A <b>and</b> r.m.s. current = 1.8 A                  | <b>A1</b> |
| 10(b)(ii) | peak current = 2.0 A <b>and</b> r.m.s. current = 2.0 A                  | <b>A1</b> |
| 10(c)(i)  | $k = 2\pi f$                                                            | <b>C1</b> |
|           | $= 2\pi \times 50$                                                      | <b>A1</b> |
|           | $= 310 \text{ rad s}^{-1}$                                              |           |
| 10(c)(ii) | power = $V_{\text{RMS}}^2 / R$ <b>or</b> power = $V_0^2 / 2R$           | <b>C1</b> |
|           | $R = (240 / \sqrt{2})^2 / 3200$ <b>or</b> $R = 240^2 / (2 \times 3200)$ | <b>A1</b> |
|           | $R = 9.0 \, \Omega$                                                     |           |